

Table of Contents

iMotions Lab > Analysis

Using Annotations 2

Using Annotations

How to Create Annotations to Analyze Data in iMotions

Annotations are an essential tool for analyzing data in iMotions, enabling you to mark time periods relevant for data analysis. Once annotations are created, Signal Processing results and AOI metrics can be calculated for these time periods.

Annotations are versatile tool that can be applied to a stimulus (Stimulus Annotations) or recorded data (Respondent Annotations). Respondent Annotations can be automatically created (Automated Annotations) based on [event markers](#) in recordings. This includes [Live Markers](#), [API event markers](#), [Analysis events](#), or any event markers created/available in the [Replay](#) (e.g. manually-defined events, standard iMotions events, respondent input events, analysis events, etc.).

[Additional information about iMotions Annotations.](#)

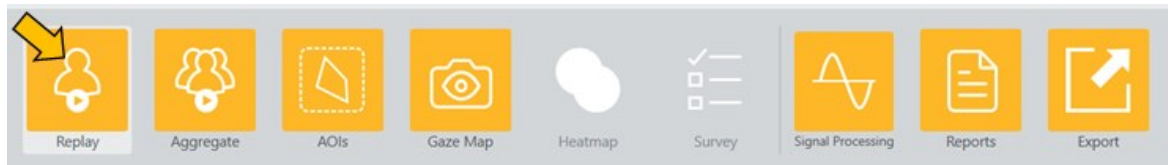
Types of Annotations

In iMotions, you can create Stimulus Annotations, Respondent Annotations, and Automated Annotations.

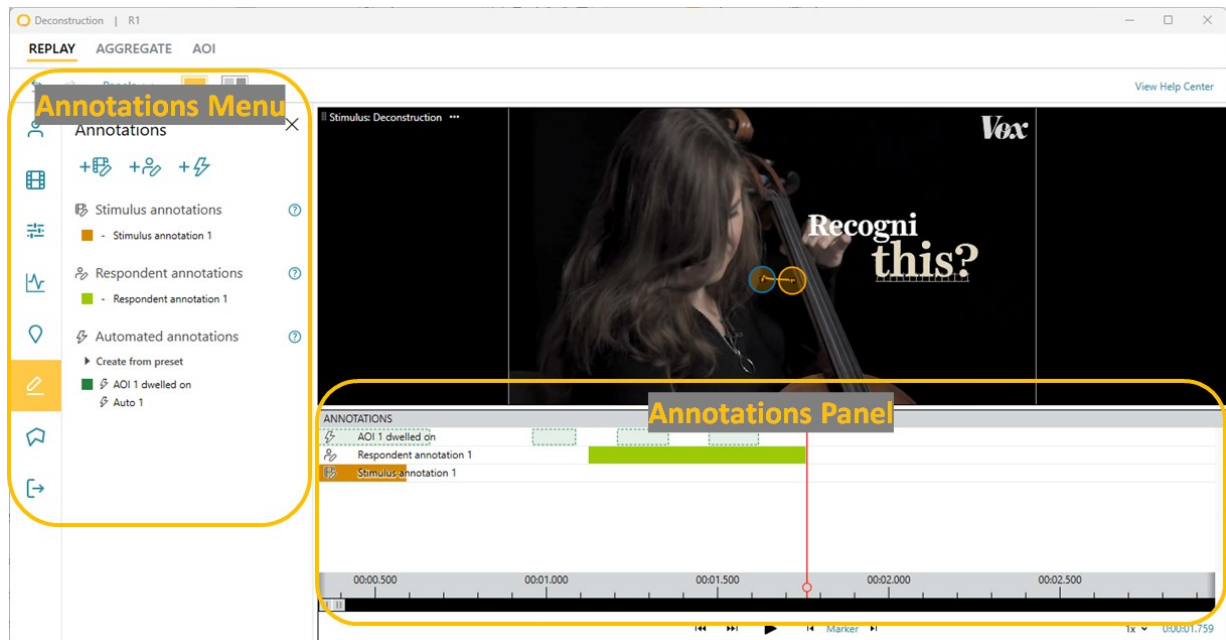
- Stimulus Annotations mark time periods within a *static* stimulus (consistent duration/content across respondents) and are applied to all data recorded with annotated stimuli. Stimulus annotations are generally used to mark relevant time periods within video or image stimuli. Stimulus Annotations must be defined manually.
- Respondent Annotations mark time periods within individual respondent recordings and unique to each respondent. Respondent Annotations can be defined manually or automatically.
- Automated Annotations also mark time periods within individual respondent recordings, but are automatically applied based on event markers in the recordings (available in iMotions 9.2 or newer). Start/End triggers can be selected from any Event Markers available in your study.

Creating Annotations

Once a respondent is recorded, Annotations can be created and edited in the Replay window. The Aggregate and AOI analysis windows also allow you to create and edit Annotations. Annotations appear in the Gaze Mapping editor, where they can define timelines for the Gaze Mapping procedure, but they cannot be edited here.



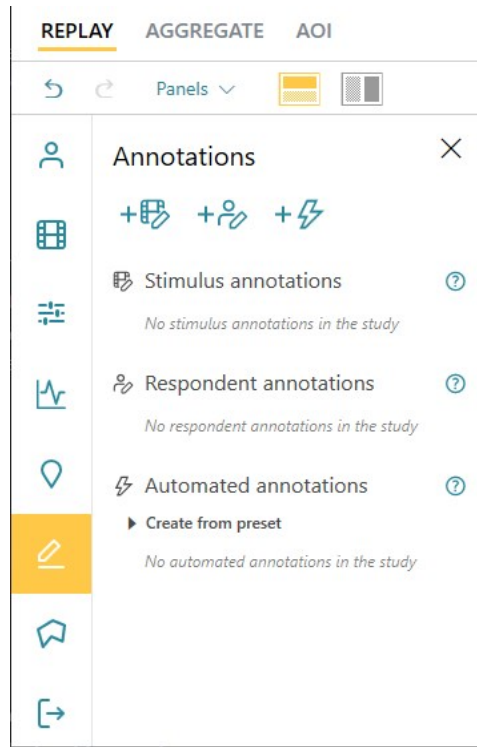
Open Replay to find Annotations Menus/Panel



Replay Window with Annotations Menu (left) and Annotations Panel (bottom right)

Annotations Menu

The Annotations Menu shows a list of existing Annotations. This menu is used to (1) create and (2) edit properties for existing Annotations. You can open the Annotations Menu by selecting the icon in the left-hand menu. New Annotations are created with the buttons at the top of this menu. Existing Annotations are categorized and listed below.

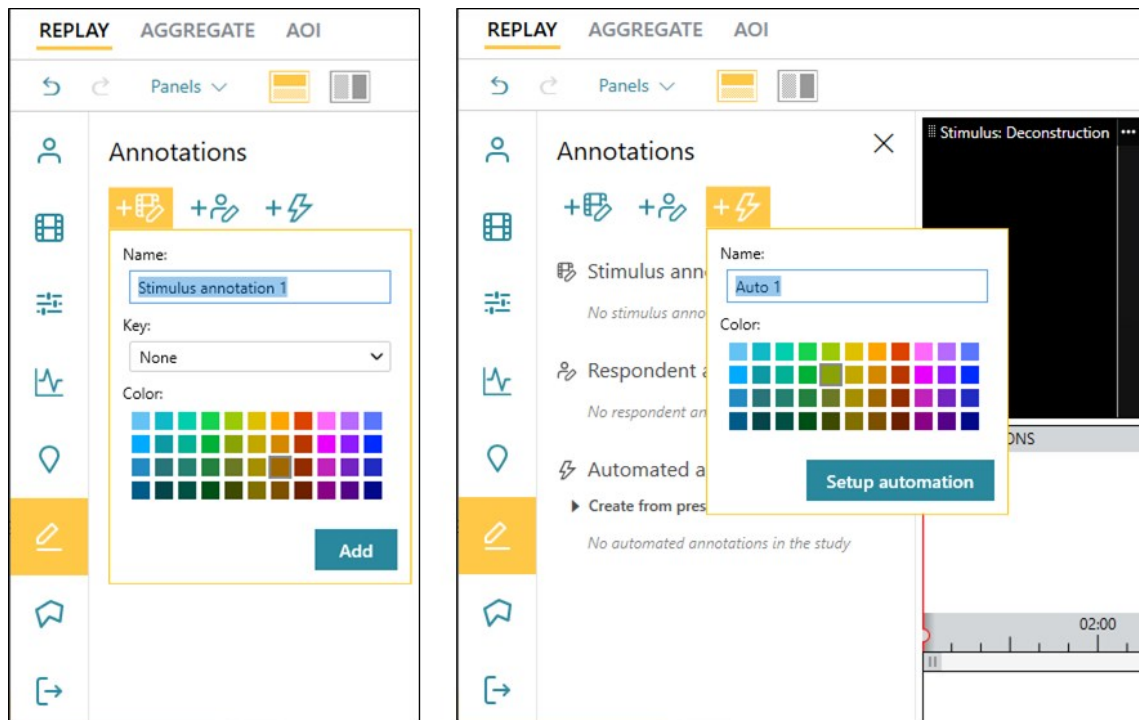


Annotations Menu

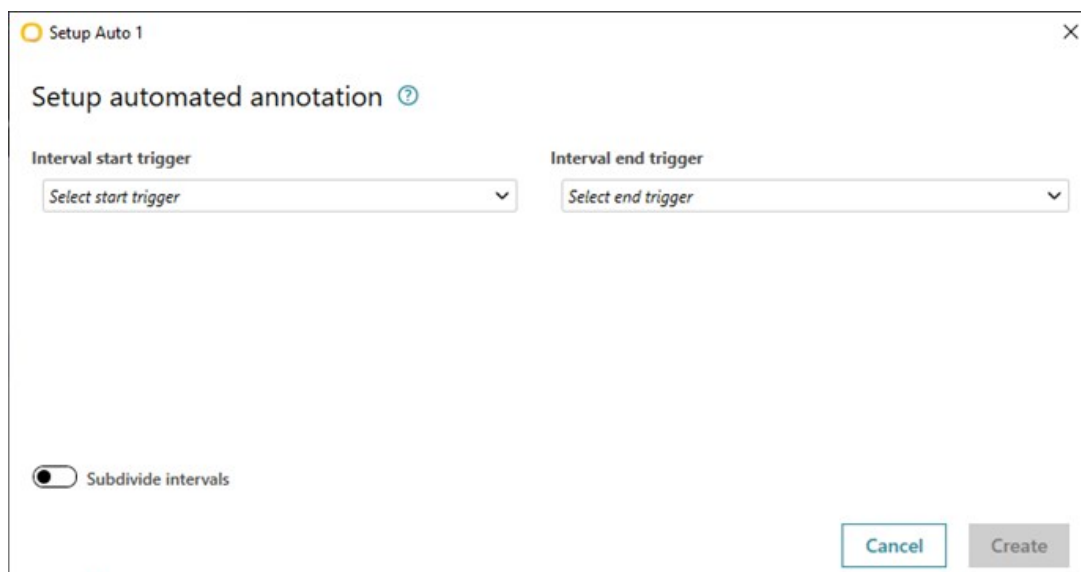
Create an Annotation

Clicking the Stimulus Annotation or Respondent Annotation buttons at the top will open the panel for naming the annotation, selecting a keyboard key (used when defining annotations), and the color (displayed in the Annotations Panel). Stimulus and Respondent Annotations are then manually defined in the Annotations Panel or imported from a CSV file (described below).

Clicking the Automated Annotation button also opens a panel for name and color. Clicking the 'Setup annotation' button opens the pop-up window for defining the automated annotation by selecting start and end triggers from event markers.



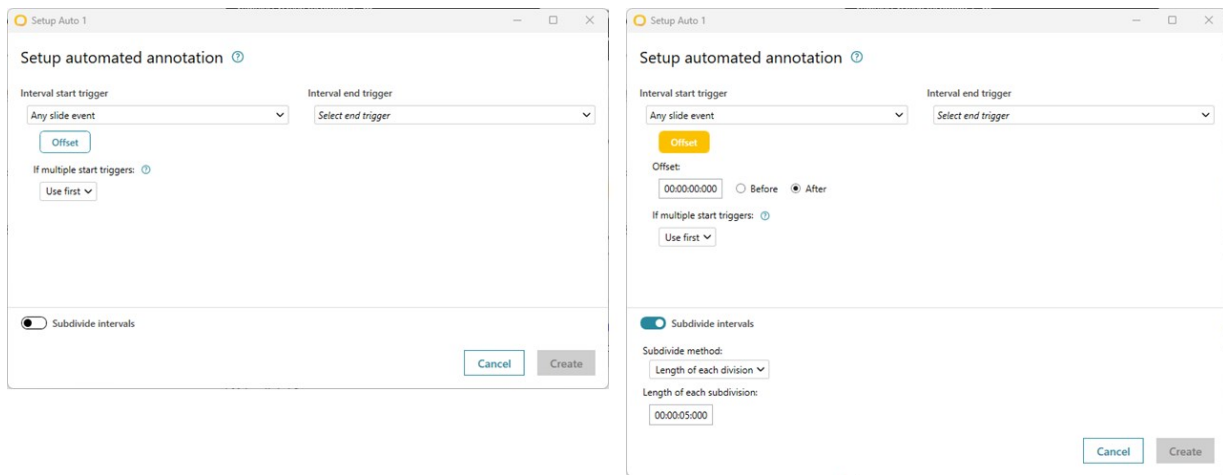
Creating a new annotation



Setting up automated annotations

Setup Automated Annotation

'Offset' controls any shift in the start or end of annotations periods. 'Multiple start/end triggers' controls how automated annotations are created when multiple triggers are encountered without a corresponding end/start. 'Subdivide intervals' allows further sub-division of automated annotations and select the 'Length of each division' or 'Number of divisions'.

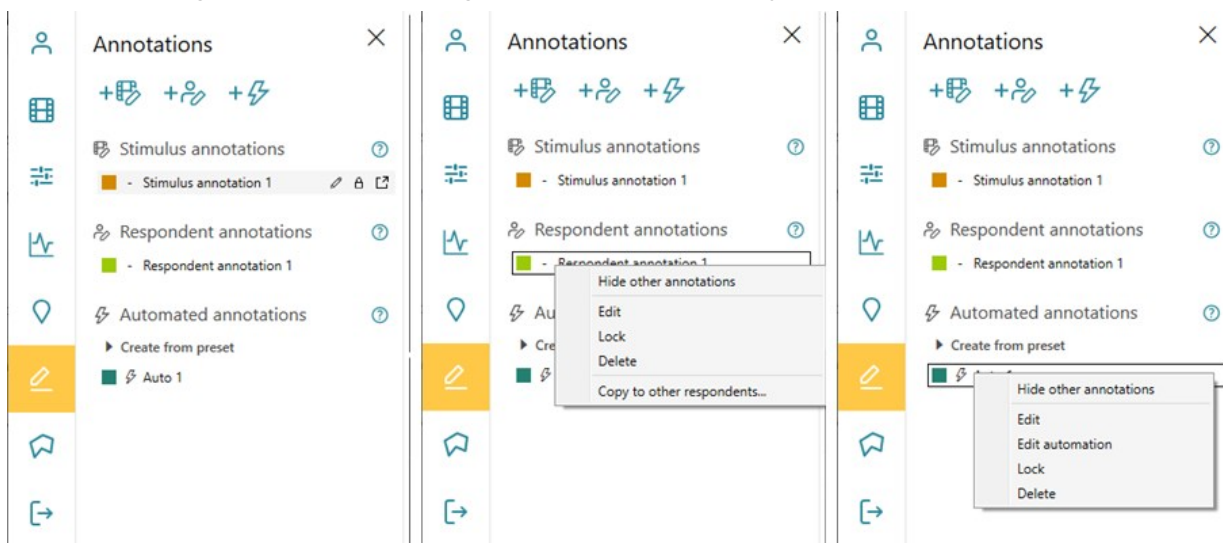


Automated Annotations with Offset and Sub-interval options

Adjust Annotation Properties

Hovering over an existing annotation then clicking the pencil icon allows adjustment of Annotation properties. The lock icon can be used to lock/unlock Annotations. The pop-up icon will open a window showing Annotation start/end time and comments.

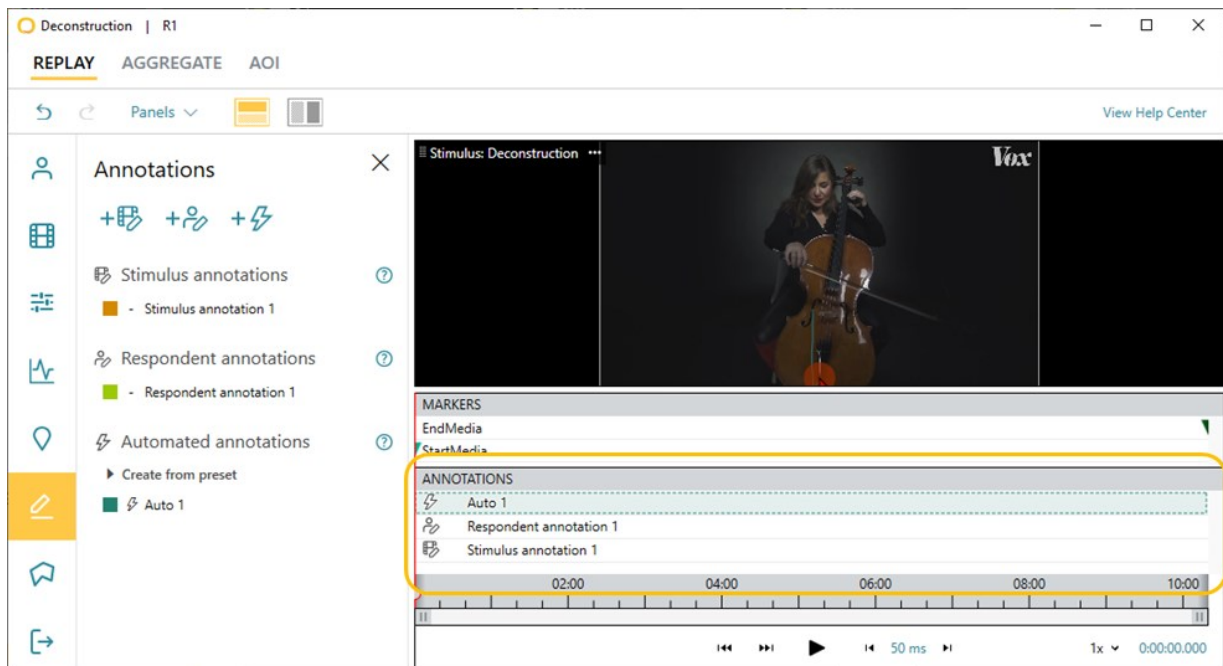
You can also right-click on an existing annotation for these adjustments.



Hovering or right-click on any Annotation for editing options

Annotations Panel (Viewing and Editing)

The Annotations Panel can be found above the timeline. Existing annotations and associated time periods are displayed here.



Annotation Panel showing existing annotations (listed in the Annotations Menu to the left)

1. Navigate to Annotations Panel

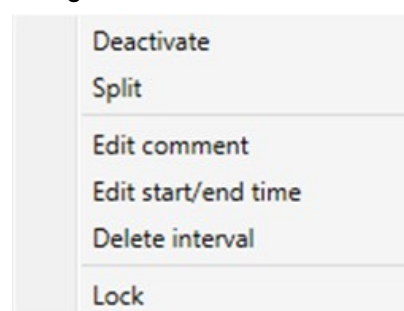
The Annotations Panel displays the timeline for each annotation. The 'Panels' drop-down menu can be used to display/hide the Annotation Panel.

2. Define manual annotations (Stimulus Annotations and Respondent Annotations)

You can use your mouse to define Stimulus and Respondent Annotations by clicking-and-dragging in the Annotations Panel. You can also use a keyboard key to enter selected points as the start/end of an annotation (according to annotations settings).

3. Add comments or adjust Annotations as needed

Comments and other adjustments are available for Stimulus and Respondent Annotations. Right-clicking on any Stimulus or Respondent Annotation will open a drop-down menu for adding comments and other changes to Annotation.



Right-click on Annotation for drop-down menu

- Deactivate - removes annotation from analysis but leaves the annotation definition
- Split - creates an additional annotation instance, splitting the selected annotation at the selected time point
- Edit comment - allows input of text comments (seen in results files and replay windows)
- Edit start/end time - allows input of specific time points for the start and end of the

annotation

- Delete interval - removes annotation instance completely
- Lock - prevents further changes to the annotation (unless unlocked)

Automated Annotations are adjusted with the 'Setup automated annotation' process

You can adjust Automated Annotations in the Annotations Menu by hovering over an existing annotation then clicking the pencil icon or right-clicking on the annotation and selecting 'Edit automation'

Event Markers and Automated Annotations

- Event Markers and Annotations are fundamental tools for analyzing data in iMotions.
- Event Markers are associated with individual time points.
- Annotations are associated with time periods, which are defined by start/end Event Markers (manually or automatically).
- Manual Annotations will generate corresponding Event Markers.
- Automated Annotations can be made from any available Event Markers (e.g. [Live Markers](#), [API event markers](#), [Analysis events](#), or any event markers created/available in the [Replay](#)).

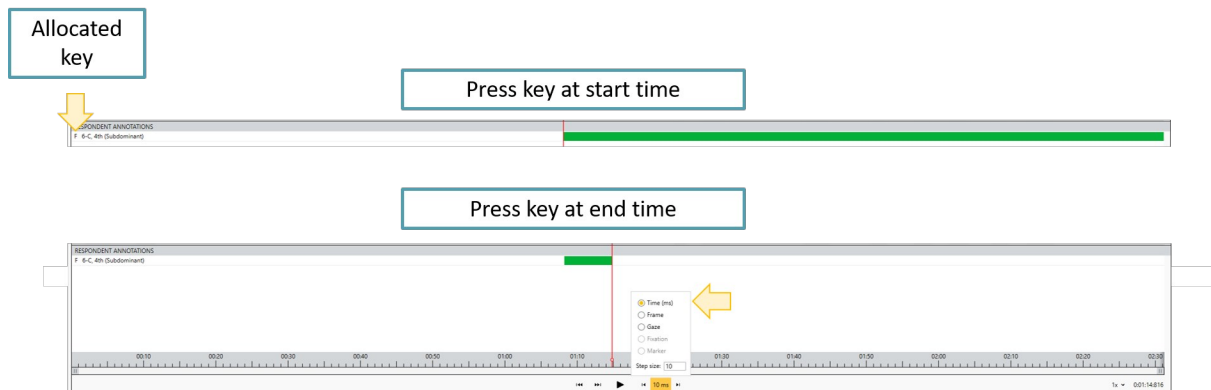
Manual Annotation Workflow

After a Manual Annotation is created, you can navigate your respondent recording to mark the relevant parts of the data on the timeline. You could either use the allocated keyboard keys (defined in your Annotation properties) or your mouse - described below:

How to place annotations using allocated keys:

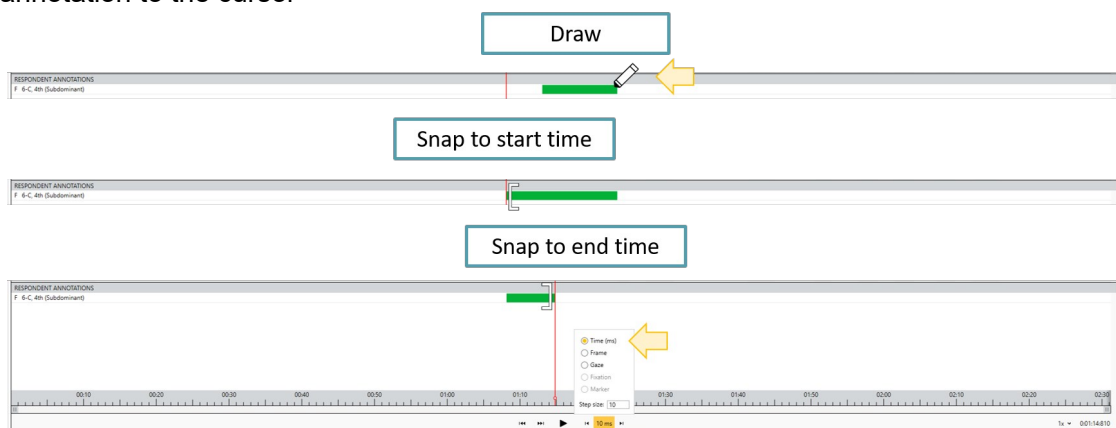
1. Pressing an allocated key will create an Annotation at the current time point in your Replay.
2. This can be done as the video plays back or by stepping through your recording more precisely:
 - a. Set the Step size for the cursor to the right level of precision (e.g., per 'x' milliseconds, per frame, etc.)
 - b. Use the left and right arrow keys to place the cursor precisely where the annotation should start.
 - c. Press the allocated key for the annotation; now, the annotation shows up from the start time until the end of the timeline.
 - d. Repeat step 2 to place the cursor precisely where the annotation should end

Press the dedicated key for the annotation; now, the annotation is set from the start time until the cursor.



How to place annotations using your mouse:

1. Hover over the Annotation panel and row for the the Annotation you want to define, which should show a Pencil icon.
2. Click the Pencil to manually draw an annotation (by clicking at the desired start location and releasing at the end location); now, the annotation is placed on the timeline.
3. To adjust the start time, first, set the Step size for the cursor to the right level of precision (e.g., per 'x' milliseconds, per frame, etc.)
4. Use the left and right arrow keys to place the cursor precisely where the annotation should start.
5. Hoover over the edges of the annotation; note that the mouse icon becomes a '[' and ']' in these areas
6. Grasp the start of the annotation (by clicking and holding the mouse '[') and snap the annotation to the cursor
7. Hereafter, repeat step 4 and 5 and place the cursor precisely where the annotation should end
8. Grasp the end of the annotation (by clicking and holding the mouse ']') and snap the annotation to the cursor



Importing Annotations from CSV file

You can also define Stimulus and Respondent Annotations by importing a CSV file. The CSV file must be formatted with a header row that contains 'Respondent Name', 'Stimulus Name', 'Marker Type', 'Marker Name', 'Start Time', 'End Time', 'Comment', and 'Color'. Note: These headers must be in this order and verbatim, at the top of the CSV file, as shown in this example

 AnnotationImportFile.csv

Respondent Name	Stimulus Name	Marker Type	Marker Name	Start Time (ms)	End Time (ms)	Comment	Color
			First 10s	0	10000	Annotation for first ten seconds of every stimulus	#0000FF
	Chewing gum ad	Stimulus Annotation	Brand mentioned	5000	5500	Actress mentions brand	#2FD09F
	Chewing gum ad	Stimulus Annotation	Brand mentioned	21000	21500	Brand presented in video	#2FD09F
Respondent 1	Chewing gum ad	Respondent Annotation	Laughed	4000	4500		#FF0000
Respondent 1	Rating scale	Respondent Annotation	Laughed	1200	1700		#FF0000
Respondent 2	Chewing gum ad	Respondent Annotation	Laughed	8500	9000		#FF0000
Respondent 2	Rating scale	Respondent Annotation	Laughed	3500	3800		#FF0000

CSV file format

Column 1: Respondent Name (Allowed values: Text input or blank)

If specified, the annotation will only be applied to the respondent with this name.

If left blank, the annotation will be applied to all respondents in the study

Column 2: Stimulus Name (Allowed values: Text input or blank)

If specified, the annotation will only be applied to the stimulus with this name.

If left blank, the annotation will be applied to all stimuli in the study.

Column 3: Marker Type (Allowed values are: 'Respondent Annotation', 'Stimulus Annotation' or blank)

If Respondent Annotation is specified, the annotation will be added as a Respondent Annotation based on the Respondent Name

If Stimulus Annotation is specified, the annotation will be added as Stimulus Annotation and appear in aggregated replays. Stimulus annotations are always added to all respondents, so specifying a

Respondent name is not needed.

If left blank, the annotation is added as Respondent annotation.

Column 4: Marker Name (Allowed values: Text input)

Name of the Annotation. This must be specified and cannot be left blank.

Column 5: Start Time (ms) (Allowed values: Floats, i.e. numeric values)

Start time of the annotation relative to stimulus start, in milliseconds.

Column 6: End Time (ms) (Allowed values: Floats, i.e. numeric values)

End time of the annotation relative to stimulus start, in milliseconds.

Column 7: Comment (Allowed values: Text input or blank)

Comment to be added to the current annotation

Column 8: Color (Allowed values are: Text input or blank.

Color must be specified in hex code (e.g. #FF0000 creates a red Annotation)

To import the CSV file

1. Right-click on the study name.
2. Select External Data > Import annotations from CSV.
3. Browse for your file and import.

Using Event Markers to Create Annotations

Automated Annotations can be created from any Event Markers in your recordings. This includes (1) iMotions Events for stimulus presentation events, keyboard events, and mouse events and (2) Analysis Events from Signal Processing (R Notebooks) and AOI analysis, such as the onset/offset of GSR peaks, blink, muscle contractions, or AOI dwells. These Analysis Markers can be used as start/end triggers for Automatic Annotations. This greatly facilitates multi-modal analysis, allowing each sensor to inform the analysis of others. Results from these Annotations provide information about the relationships between signal events.

iMotions Markers

Keyboard and Mouse events

Keyboard events - Indicates key presses and key name

Mouse events - Indicates mouse button presses, cursor movement, and scrolling

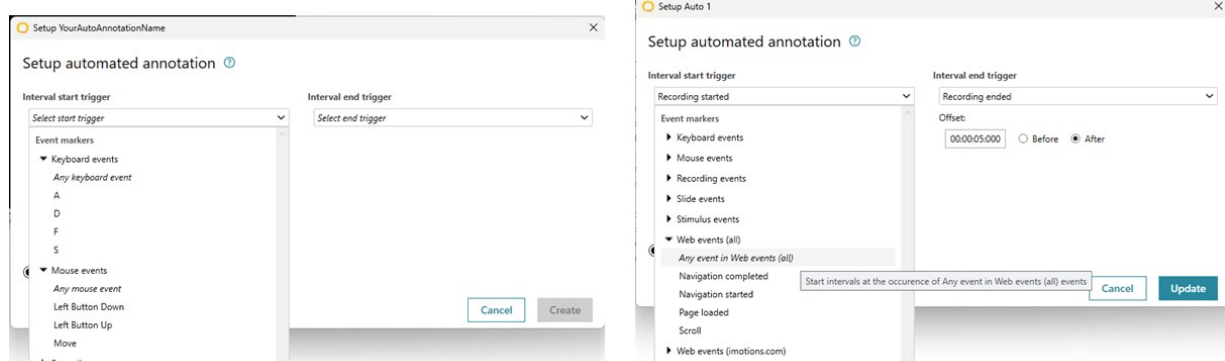
iMotions Events for stimulus presentations events

Recording events - Mark the start and end of the respondent recording (across all stimuli)

Slide events - Mark the start and end of iMotions stimulus presentation (in general)

Stimulus events - Mark the start and end of iMotions stimulus presentation (labeled with stimulus name)

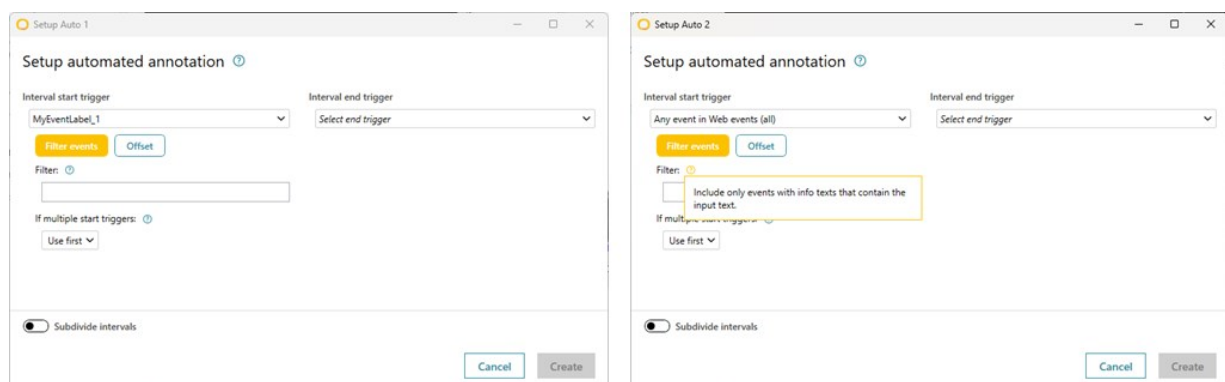
Web events - Mark web browser events when the 'Page loaded' and 'Scroll' occurred (for all or specific webpages)



Keyboard and Mouse Events (left); Other iMotions Events and Web Events (right)

Marker Filters

Events with additional properties (e.g. Web Events, API Events, or LSL Events) can be selected with a text filter, available in the automated annotation window (in iMotions 10.1.2 onward). This filter will search the event for the entered string and only retain those events that contain the string.



Event Marker Filters for Automated Annotations for web stimuli or API events

Analysis Markers

After AOI Analysis and/or Signal Process (R Notebooks)

For example, if you want to know the number of GSR peaks that occur while respondents dwelled on an AOI, you can setup and run the AOI analysis, then use the resulting AOI Dwell Start/End markers (found in the Markers Menu/Panel) to create Automated Annotations while the AOI was dwelled upon.

To Use Analysis Events to Create Automated Annotations:

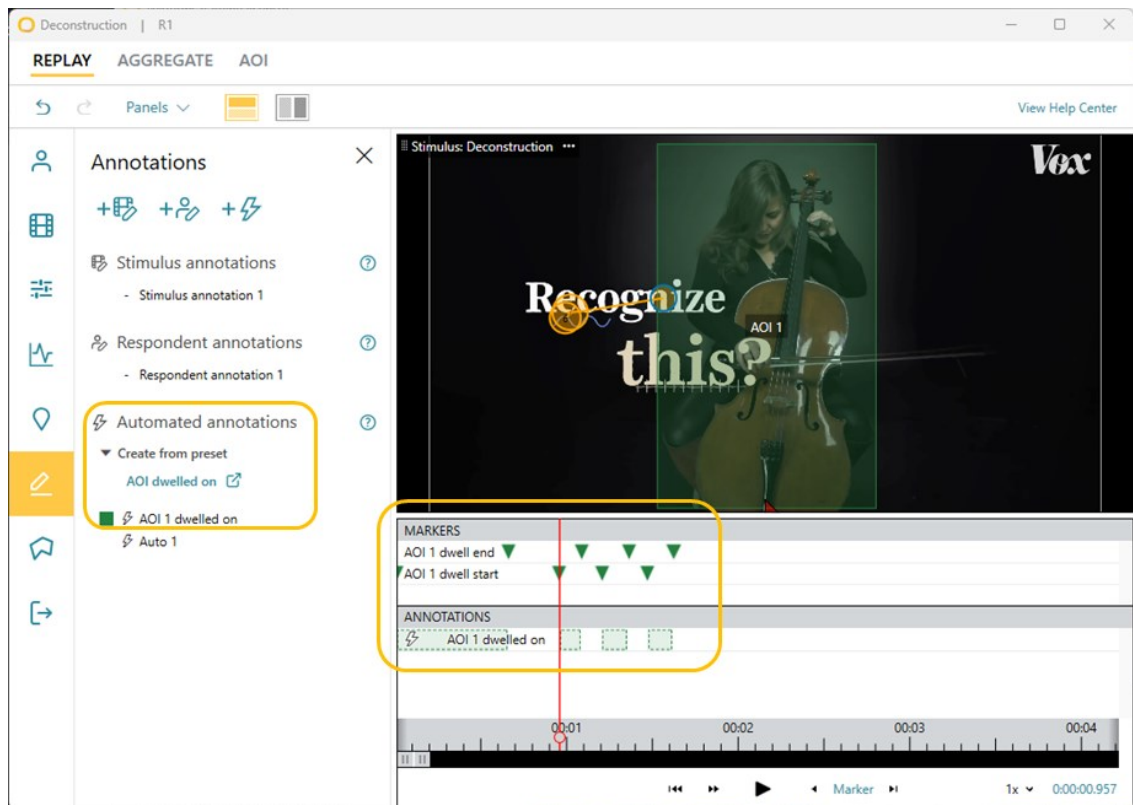
1. Set up and run AOI or Signal Processing Analysis
2. Open a Respondent Replay and open the Annotations Menu and Annotations Panel.
3. Click the Create Automated Annotation button in the Annotations Menu.
4. Select a start and end trigger from Analysis Events - typically, onset and offset pairs.

Creating Automated Annotation with Analysis event triggers (i.e. AOI 1 dwell start/end)

5. Some Analysis events can be interpreted into Automated Annotations with 'preset' options (currently available for 'AOI dwelled on').

Creating Automated Annotations for Analysis events with preset options (i.e. 'AOI dwelled on')

6. Click 'Create' to return to the Replay window where Automated Annotations appear (i.e. 'AOI 1 dwelled on').



Replay Window showing Automated Annotation for Analysis Events (i.e. 'AOI 1 dwelled on')

Exporting Annotations

You can export Annotations independently or along with Sensor Exports or R-Processed Exports.



Analysis Export button

Annotations Data Files

Annotations can be exported into a CSV file.

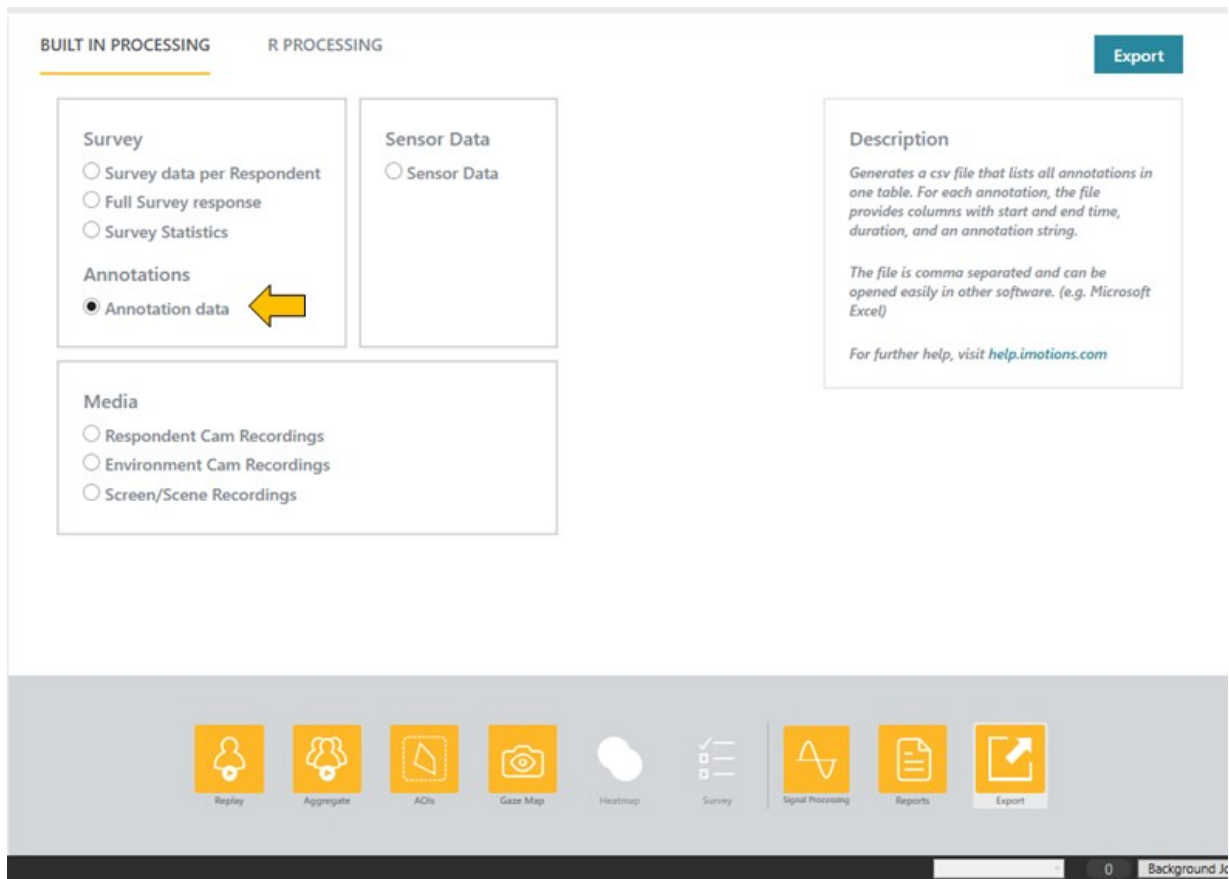
Click 'Export' to create two files:

Individual Annotation Data.csv contains a list of all annotations with the 'Respondent Name', 'Stimulus Name', 'Marker Type', 'Marker Name', 'Start Time', 'End Time', 'Comment', and 'Color'.

 Example_Individual Annotation Data.csv

Aggregate Annotation Results.csv contains the average duration for each annotation, along with the standard deviation and other information about the analysis

 Example_Aggregate Annotation Result.csv

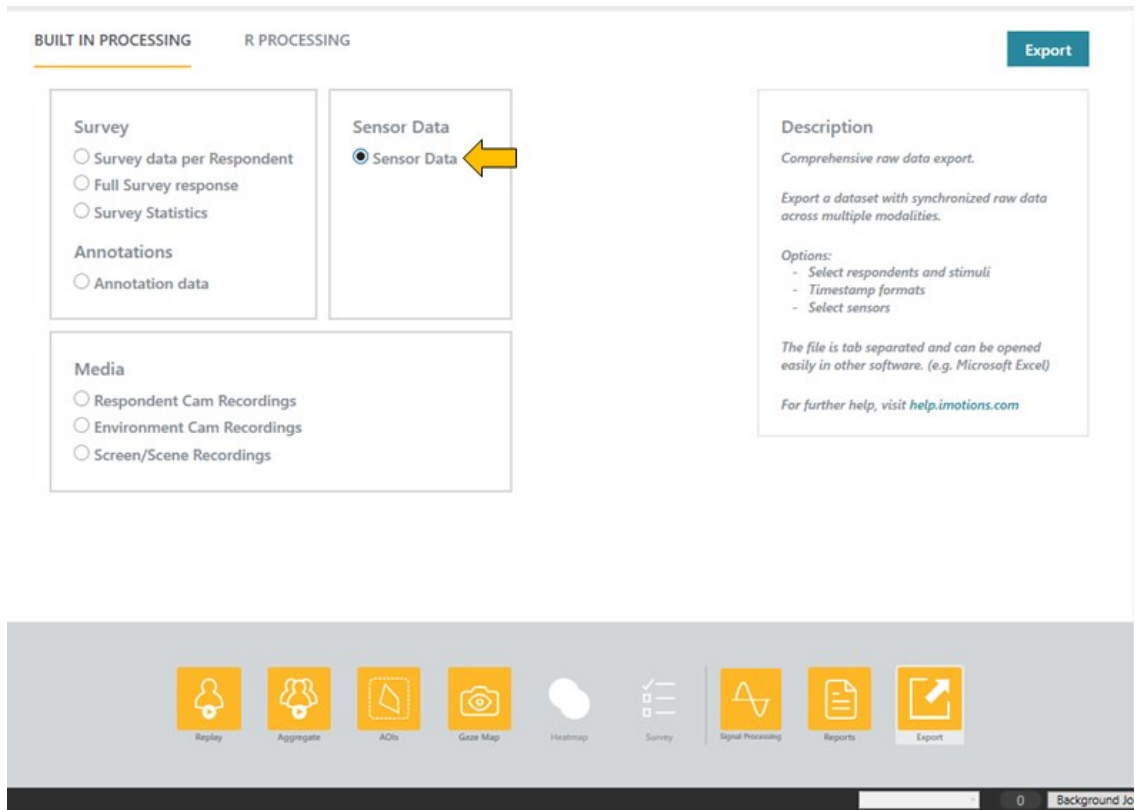


Annotation data export

Annotations in Sensor Data export files

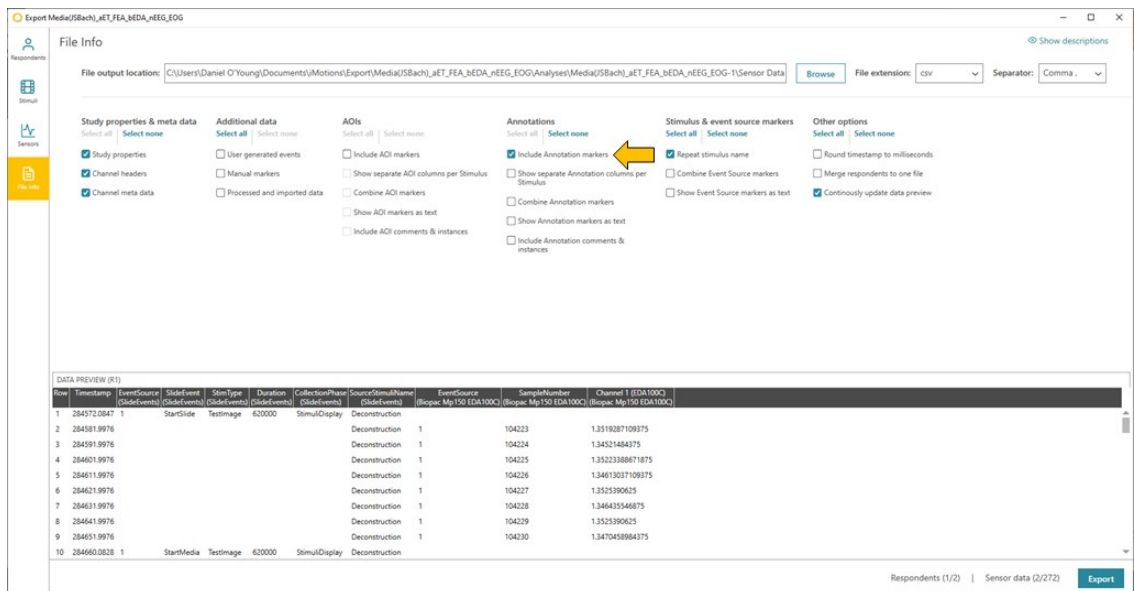
Annotations can also be exported as part of the sensor data export CSV files. Here, the annotations mark all data samples that are present within the time range of the annotations.

1. Open the 'Sensor Data' export window



Sensor Data export

2. Select the 'File Info' tab and check 'Include Annotation markers'
3. Check any additional options for Annotation markers (by default, a column is created for each Annotation with a 1 or 0 indicating if the data point is associated with the Annotation)



File Info Annotation settings for Sensor Data export

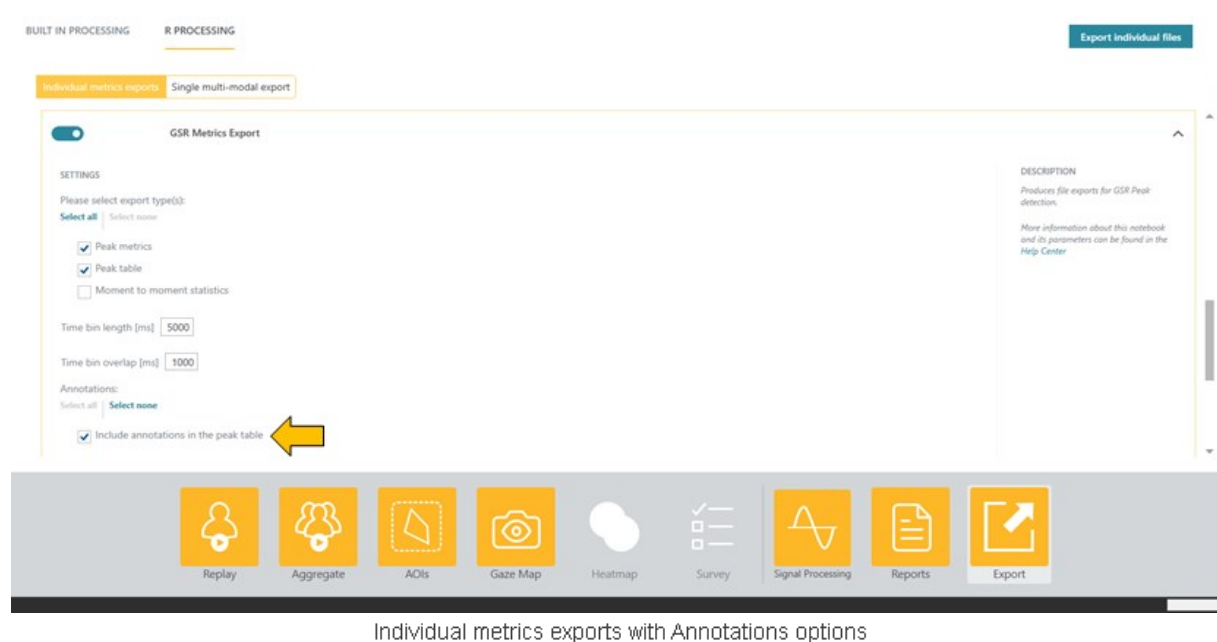
 Example_SensorDataExport_001_R1.csv

Annotations in R-Processing export files

Annotations are automatically exported as part of most R-Processing export CSV files. Here, Annotations serve as 'measurement windows' for the data generated by R-Notebooks (see also: [R-Notebooks](#)) and are used to calculate exported metrics.

For instance, if you would have applied a peak detection R-Notebook for detecting peaks in the electrodermal activity data (i.e., a measurement of emotional arousal) and you export the metric export file, then you would be able to retrieve the peak results *for every annotation* and even *for every annotation fragment* (e.g., whether a peak was detected, how many peaks were detected, the average amplitude of the peaks, etc.).

You can also include Annotations for 'Individual metrics exports', where Annotation markers are included where applicable.



 Example_GSRPeakMetrics.csv

 Example_GSRPeakTable.csv

Scenarios for Using Annotations

Annotations are a versatile tool for analyzing data in iMotions - by classifying time periods for subsequent comparisons. Below, we discuss how Annotations apply to common research scenarios and provide recommendations for types of Annotations to use in different cases.

- Stimulus Coding (Segmenting stimuli)
- Human Behavior Coding (Qualitative segmentation by experimenter)
- Event Analysis (Epoching data with event markers)
- Interval Analysis (Time-window binning)
- Using Sensor Analysis Events to Create Annotations

Stimulus Coding (Segmenting stimuli)

You can segment stimuli into relevant time periods for your analysis with Manual or Automated Annotations.

Stimulus Annotations are recommended for coding static stimuli (that have consistent duration and content across respondents, such as images and videos). Any Replay window can be used to create Stimulus Annotations manually, then applied them across all recordings involving the stimulus in the study.

Stimulus Annotations are typically used for analysis of stimulus-driven events, like reactions to video commercials, trailers, films, etc. Hence, if your analysis focuses on the respondents' emotional response to specific time periods in a the video, we recommend Stimulus Annotations. This has two benefits: (1) Stimulus Annotation automatically apply to all respondents and (2) Aggregate Analyses are available for Stimulus Annotations (e.g. heatmaps and group metrics for observing responses from all respondents).

Respondent Annotations are necessary for dynamic stimuli, due to their variable duration and content, which are unique to each respondent. This includes screen recordings, websites, cameras, VR environments, etc. These Annotations can be applied manually or automatically based on available event markers.

Automated Respondent Annotations are recommended when dynamic stimuli must be segmented into more numerous time periods. This requires creating Event Markers that indicate the start and end of each Annotation. This can be done with (1) Live Markers created while monitoring data collection, (2) Manual Respondent Markers created while reviewing stimulus recordings, or (3) API Markers created by stimulus presentation software (additional setup for iMotions Event Receiving API required), which are covered in the other scenarios described here.

For Stimulus Coding (Video media):

1. Open a Respondent Replay and open the Annotations Menu and Annotations Panel.
2. Click the Create Stimulus Annotation button in the Annotation Menu and label as needed.
3. Set a Key for each annotation, which will be used to create the start/end points of the Annotations.
4. Navigate to the start point for an Annotation by while playing back your recording (additional playback speeds available: 0.25x, 0.5x, 2x) or by selecting the time point with the red cursor in the replay timeline.
5. Press the annotation key to create the start of an Annotation.
6. Navigate to the end point for the Annotation and press the annotation key again to end the Annotation
7. Adjust Annotations (as needed) in the Annotation Panel, by clicking and dragging the start or end point. This Annotation will snap to the current timepoint (indicated by the red cursor)
8. These steps can be used to define multiple instances of any Annotation (unless dragged further from the red cursor).
9. For precise Replay navigation, you can pause the recording and define a Step for moving through the recording and defining Annotations (option to set each Step to iterate by milliseconds, number of frames, or marked events). Hereafter, forward/backward buttons (in the Replay window) or the left/right arrow keys (on the keyboard) will step through the replay at the selected interval.

Demo study: Sony Bravia (Testing of video commercials)

Human Behavior Coding (Qualitative segmentation by experimenter)

Respondent Annotations can be used to code behaviors or apply any other classification to your iMotions recording (e.g. experimental conditions, factors, etc.). This allows the traditional approach of coding data based on your observations in iMotions recordings. This also allows coding based on any other information available in the iMotions Replay window, such as sensor data and analyses.

For Human Behavior Coding:

1. Open a Respondent Replay and open the Annotations Menu and Annotations Panel.
2. Click the Create Respondent Annotation button in the Annotation Menu to create an Annotation for each classification you intend to use.
3. Set a Key for each annotation, which will be used to create the start/end points of the Annotations.
4. Navigate to the start point for an Annotation by while playing back your recording (additional playback speeds available: 0.25x, 0.5x, 2x) or by selecting the time point with the red cursor in the replay timeline.
5. Press the annotation key to create the start of an Annotation.
6. Navigate to the end point for the Annotation and press the annotation key again to end the Annotation
7. Adjust Annotations (as needed) in the Annotation Panel, by clicking and dragging the start or end point. This Annotation will snap to the current timepoint (indicated by the red cursor)
8. These steps can be used to define multiple instances of any Annotation (unless dragged further from the red cursor).
9. For precise Replay navigation, you can pause the recording and define a Step for moving through the recording and defining Annotations (option to set each Step to iterate by milliseconds, number of frames, or marked events). Hereafter, forward/backward buttons (in the Replay window) or the left/right arrow keys (on the keyboard) will step through the replay at the selected interval.

Note:

Multiple instances of can be defined for any Annotation.

Event Markers can be used with Automated Annotations to facilitate Human Behavior Coding.

Demo study: In-store VR (Shelf-testing in VR environment)

Event Analysis (Epoching data with event markers)

Automated Annotations can be used to define event-related 'epochs' to analyze response before, during, or after certain events. In this scenario, many repetitions of these events may occur, and you would define a time-window of a certain window length centered around the events.

For instance, it could be that you would define time-windows that range from 1 second before until 2 seconds after every event. It could also be that you would define time-windows that range from an onset event (e.g., a stimulus appears) to an offset event (e.g., a correct or incorrect response is detected) to analyze data from categorized time periods.

The precise timing of events is critical for these analyses. Ideally, Automated Annotations can be based on automatically generated Event Marker, such as include Stimulus events, Keyboard events, Mouse events, or API events. The following steps include a screenshot example that will create Automated Annotations that will start 1 second before a 'False' response/event marker and end 2 seconds after the event.

For Event Analysis:

1. Confirm that relevant Event Markers are available in your Respondent Recording.
2. Open a Respondent Replay and open the Annotations Menu and Annotations Panel.
3. Click the Create Automated Annotation button in the Annotations Menu.
4. Use the 'Setup automated annotation' interface to select relevant start and end triggers (i.e. Event Markers) for your epochs.
5. Adjust offsets (as needed) entering an offset time then selecting 'Before' or 'After' (to shift the start or end of the Annotation, accordingly).

The screenshot shows a dialog box titled "Setup False response" with a close button (X) in the top right corner. The main heading is "Setup automated annotation" with a help icon (?). The dialog is divided into two columns: "Interval start trigger" and "Interval end trigger".

Interval start trigger: A dropdown menu is set to "False". Below it, the "Offset:" is "00:00:01:000" with radio buttons for "Before" (selected) and "After". Below that, "Multiple start triggers:" has a "Use first" dropdown.

Interval end trigger: A dropdown menu is set to "Start trigger". Below it, the "Offset:" is "00:00:02:000" with radio buttons for "Before" and "After" (selected).

At the bottom left, there is a toggle switch for "Subdivide intervals" which is currently turned off. At the bottom right, there are "Cancel" and "Create" buttons.

Demo study: Strooptest (Psychology experiment)

Time Interval Analysis (Time-window binning)

Automated Annotations can also be used to bin, or sub-divide, your recordings into time-windows. You can select the (a) number or (b) duration of time windows used to define Automated Annotations. This is typically applied when time is an experimental factor (e.g. comparing start, middle, end, time-on-task) or analysis involves regular time periods (e.g. every second, minute, etc.).

For Interval Analysis:

Creating Annotations intervals

1. Open a Respondent Replay and open the Annotations Menu and Annotations Panel.
2. Click the Create Automated Annotation button in the Annotations Menu.
3. Select a start trigger from where you want to begin binning your data (e.g. Start of the Recording, Stimulus, or any other event)
4. Select an end trigger where bins will end (e.g. the End of the Recording, Stimulus, or any other event)
5. Enable 'Subdivide intervals'
6. Choose to subdivide this Annotation by 'Length of each division' or 'Number of divisions' (with equal duration) and adjust this value as needed.

Demo study: Sony Bravia (Testing of video commercials)

